Witt group: from quadratic spaces to modular categories and VOAs

Alexei Davydov

joint project with
M. Müger, D. Nikshych, and V. Ostrik

Conformal field theories and tensor categories, Beijing,
13-17 June, 2011

Quadratic vector spaces

Quadratic vector spaces

Let V be a finite dimensional vector space over a field F

Quadratic vector spaces

Let V be a finite dimensional vector space over a field F
A function $q : V \to F$ is *quadratic* if

Quadratic vector spaces

Let V be a finite dimensional vector space over a field F
A function $q : V \to F$ is *quadratic* if $q(-x) = q(x)$ and

Quadratic vector spaces

Let V be a finite dimensional vector space over a field F
A function $q : V \to F$ is *quadratic* if $q(-x) = q(x)$ and $(x, y) = q(x + y) - q(x) - q(y)$ is bilinear.

Quadratic vector spaces

Let V be a finite dimensional vector space over a field F
A function $q : V \to F$ is *quadratic* if $q(-x) = q(x)$ and $(x, y) = q(x + y) - q(x) - q(y)$ is bilinear.
A linear map $f : V \to U$ is *isometry* if

Quadratic vector spaces

Let V be a finite dimensional vector space over a field F
A function $q : V \to F$ is *quadratic* if $q(-x) = q(x)$ and $(x, y) = q(x + y) - q(x) - q(y)$ is bilinear.
A linear map $f : V \to U$ is *isometry* if $q_U(f(v)) = q_V(v)$.

Quadratic vector spaces

Let V be a finite dimensional vector space over a field F
A function $q : V \to F$ is *quadratic* if $q(-x) = q(x)$ and $(x, y) = q(x + y) - q(x) - q(y)$ is bilinear.
A linear map $f : V \to U$ is *isometry* if $q_U(f(v)) = q_V(v)$.
Direct sum: $(V, q_V) \oplus (U, q_U) = (V \oplus U, q_V \oplus q_U)$.

Quadratic vector spaces

Let V be a finite dimensional vector space over a field F
A function $q : V \to F$ is *quadratic* if $q(-x) = q(x)$ and $(x, y) = q(x + y) - q(x) - q(y)$ is bilinear.
A linear map $f : V \to U$ is *isometry* if $q_U(f(v)) = q_V(v)$.
Direct sum: $(V, q_V) \oplus (U, q_U) = (V \oplus U, q_V \oplus q_U)$.
We say that (V, q) is *non-degenerate* if

Quadratic vector spaces

Let V be a finite dimensional vector space over a field F
A function $q : V \to F$ is *quadratic* if $q(-x) = q(x)$ and $(x, y) = q(x + y) - q(x) - q(y)$ is bilinear.
A linear map $f : V \to U$ is *isometry* if $q_U(f(v)) = q_V(v)$.
Direct sum: $(V, q_V) \oplus (U, q_U) = (V \oplus U, q_V \oplus q_U)$.
We say that (V, q) is *non-degenerate* if $(\, , \,)$ is.

Quadratic vector spaces

Let V be a finite dimensional vector space over a field F
A function $q : V \to F$ is *quadratic* if $q(-x) = q(x)$ and $(x, y) = q(x + y) - q(x) - q(y)$ is bilinear.
A linear map $f : V \to U$ is *isometry* if $q_U(f(v)) = q_V(v)$.
Direct sum: $(V, q_V) \oplus (U, q_U) = (V \oplus U, q_V \oplus q_U)$.
We say that (V, q) is *non-degenerate* if $(\, , \,)$ is.
For $L \subset V$ define $L^{\perp} = \{v \in V | \, (v, L) = 0\}$

Quadratic vector spaces

Let V be a finite dimensional vector space over a field F
A function $q : V \to F$ is *quadratic* if $q(-x) = q(x)$ and
$(x, y) = q(x + y) - q(x) - q(y)$ is bilinear.
A linear map $f : V \to U$ is *isometry* if $q_U(f(v)) = q_V(v)$.
Direct sum: $(V, q_V) \oplus (U, q_U) = (V \oplus U, q_V \oplus q_U)$.
We say that (V, q) is *non-degenerate* if $(\, , \,)$ is.
For $L \subset V$ define $L^\perp = \{v \in V | \, (v, L) = 0\}$
L is *isotropic* if

Quadratic vector spaces

Let V be a finite dimensional vector space over a field F
A function $q : V \to F$ is *quadratic* if $q(-x) = q(x)$ and
$(x, y) = q(x + y) - q(x) - q(y)$ is bilinear.
A linear map $f : V \to U$ is *isometry* if $q_U(f(v)) = q_V(v)$.
Direct sum: $(V, q_V) \oplus (U, q_U) = (V \oplus U, q_V \oplus q_U)$.
We say that (V, q) is *non-degenerate* if $(\, , \,)$ is.
For $L \subset V$ define $L^\perp = \{v \in V | \, (v, L) = 0\}$
L is *isotropic* if $L \subset L^\perp, \quad q|_L = 0$.

Quadratic vector spaces

Let V be a finite dimensional vector space over a field F
A function $q : V \to F$ is *quadratic* if $q(-x) = q(x)$ and
$(x, y) = q(x + y) - q(x) - q(y)$ is bilinear.
A linear map $f : V \to U$ is *isometry* if $q_U(f(v)) = q_V(v)$.
Direct sum: $(V, q_V) \oplus (U, q_U) = (V \oplus U, q_V \oplus q_U)$.
We say that (V, q) is *non-degenerate* if $(\, , \,)$ is.
For $L \subset V$ define $L^\perp = \{v \in V | \, (v, L) = 0\}$
L is *isotropic* if $L \subset L^\perp, \quad q|_L = 0$.
L is *Lagrangian* if

Quadratic vector spaces

Let V be a finite dimensional vector space over a field F
A function $q : V \to F$ is *quadratic* if $q(-x) = q(x)$ and
$(x, y) = q(x + y) - q(x) - q(y)$ is bilinear.
A linear map $f : V \to U$ is *isometry* if $q_U(f(v)) = q_V(v)$.
Direct sum: $(V, q_V) \oplus (U, q_U) = (V \oplus U, q_V \oplus q_U)$.
We say that (V, q) is *non-degenerate* if $(\, , \,)$ is.
For $L \subset V$ define $L^\perp = \{v \in V | \, (v, L) = 0\}$
L is *isotropic* if $L \subset L^\perp, \quad q|_L = 0$.
L is *Lagrangian* if $L = L^\perp, \quad q|_L = 0$.

Quadratic vector spaces

Let V be a finite dimensional vector space over a field F
A function $q : V \to F$ is *quadratic* if $q(-x) = q(x)$ and
$(x, y) = q(x + y) - q(x) - q(y)$ is bilinear.
A linear map $f : V \to U$ is *isometry* if $q_U(f(v)) = q_V(v)$.
Direct sum: $(V, q_V) \oplus (U, q_U) = (V \oplus U, q_V \oplus q_U)$.
We say that (V, q) is *non-degenerate* if $(\, , \,)$ is.
For $L \subset V$ define $L^\perp = \{v \in V | \, (v, L) = 0\}$
L is *isotropic* if $L \subset L^\perp, \quad q|_L = 0$.
L is *Lagrangian* if $L = L^\perp, \quad q|_L = 0$.
(H, q) is *hyperbolic* if

Quadratic vector spaces

Let V be a finite dimensional vector space over a field F
A function $q : V \to F$ is *quadratic* if $q(-x) = q(x)$ and
$(x, y) = q(x + y) - q(x) - q(y)$ is bilinear.
A linear map $f : V \to U$ is *isometry* if $q_U(f(v)) = q_V(v)$.
Direct sum: $(V, q_V) \oplus (U, q_U) = (V \oplus U, q_V \oplus q_U)$.
We say that (V, q) is *non-degenerate* if $(\, , \,)$ is.
For $L \subset V$ define $L^\perp = \{v \in V | \, (v, L) = 0\}$
L is *isotropic* if $L \subset L^\perp, \quad q|_L = 0$.
L is *Lagrangian* if $L = L^\perp, \quad q|_L = 0$.
(H, q) is *hyperbolic* if $\exists \, L \subset H$ Lagrangian.

Quadratic vector spaces

Let V be a finite dimensional vector space over a field F
A function $q : V \to F$ is *quadratic* if $q(-x) = q(x)$ and
$(x, y) = q(x + y) - q(x) - q(y)$ is bilinear.
A linear map $f : V \to U$ is *isometry* if $q_U(f(v)) = q_V(v)$.
Direct sum: $(V, q_V) \oplus (U, q_U) = (V \oplus U, q_V \oplus q_U)$.
We say that (V, q) is *non-degenerate* if $(\, , \,)$ is.
For $L \subset V$ define $L^\perp = \{v \in V | \, (v, L) = 0\}$
L is *isotropic* if $L \subset L^\perp, \quad q|_L = 0$.
L is *Lagrangian* if $L = L^\perp, \quad q|_L = 0$.
(H, q) is *hyperbolic* if $\exists \, L \subset H$ Lagrangian.
$(V, q) \oplus (V, -q)$ is hyperbolic, with Lagrangian
$L = \{(v, v) | \, v \in V\}$.

Quadratic vector spaces

Let V be a finite dimensional vector space over a field F
A function $q : V \to F$ is *quadratic* if $q(-x) = q(x)$ and
$(x, y) = q(x + y) - q(x) - q(y)$ is bilinear.
A linear map $f : V \to U$ is *isometry* if $q_U(f(v)) = q_V(v)$.
Direct sum: $(V, q_V) \oplus (U, q_U) = (V \oplus U, q_V \oplus q_U)$.
We say that (V, q) is *non-degenerate* if $(\, , \,)$ is.
For $L \subset V$ define $L^\perp = \{v \in V | \, (v, L) = 0\}$
L is *isotropic* if $L \subset L^\perp, \quad q|_L = 0$.
L is *Lagrangian* if $L = L^\perp, \quad q|_L = 0$.
(H, q) is *hyperbolic* if $\exists \, L \subset H$ Lagrangian.
$(V, q) \oplus (V, -q)$ is hyperbolic, with Lagrangian
$L = \{(v, v) | \, v \in V\}$.
Witt equivalence: $(V, q_V) \sim (U, q_U)$ if

Quadratic vector spaces

Let V be a finite dimensional vector space over a field F
A function $q : V \to F$ is *quadratic* if $q(-x) = q(x)$ and
$(x, y) = q(x + y) - q(x) - q(y)$ is bilinear.
A linear map $f : V \to U$ is *isometry* if $q_U(f(v)) = q_V(v)$.
Direct sum: $(V, q_V) \oplus (U, q_U) = (V \oplus U, q_V \oplus q_U)$.
We say that (V, q) is *non-degenerate* if $(\, , \,)$ is.
For $L \subset V$ define $L^\perp = \{v \in V | \, (v, L) = 0\}$
L is *isotropic* if $L \subset L^\perp, \quad q|_L = 0$.
L is *Lagrangian* if $L = L^\perp, \quad q|_L = 0$.
(H, q) is *hyperbolic* if $\exists \, L \subset H$ Lagrangian.
$(V, q) \oplus (V, -q)$ is hyperbolic, with Lagrangian
$L = \{(v, v) | \, v \in V\}$.
Witt equivalence: $(V, q_V) \sim (U, q_U)$ if $\exists \, (H_i, q_i)$ hyperbolic
$(V, q_V) \oplus (H_1, q_1) \cong (U, q_U) \oplus (H_i, q_i)$ an isometry.

Quadratic vector spaces

Let V be a finite dimensional vector space over a field F
A function $q : V \to F$ is *quadratic* if $q(-x) = q(x)$ and
$(x, y) = q(x + y) - q(x) - q(y)$ is bilinear.
A linear map $f : V \to U$ is *isometry* if $q_U(f(v)) = q_V(v)$.
Direct sum: $(V, q_V) \oplus (U, q_U) = (V \oplus U, q_V \oplus q_U)$.
We say that (V, q) is *non-degenerate* if $(\, , \,)$ is.
For $L \subset V$ define $L^\perp = \{v \in V | \, (v, L) = 0\}$
L is *isotropic* if $L \subset L^\perp, \quad q|_L = 0$.
L is *Lagrangian* if $L = L^\perp, \quad q|_L = 0$.
(H, q) is *hyperbolic* if $\exists \, L \subset H$ Lagrangian.
$(V, q) \oplus (V, -q)$ is hyperbolic, with Lagrangian
$L = \{(v, v) | \, v \in V\}$.
Witt equivalence: $(V, q_V) \sim (U, q_U)$ if $\exists \, (H_i, q_i)$ hyperbolic
$(V, q_V) \oplus (H_1, q_1) \cong (U, q_U) \oplus (H_i, q_i)$ an isometry.
Witt group $W(F)$

Quadratic vector spaces

Let V be a finite dimensional vector space over a field F
A function $q : V \to F$ is *quadratic* if $q(-x) = q(x)$ and
$(x, y) = q(x + y) - q(x) - q(y)$ is bilinear.
A linear map $f : V \to U$ is *isometry* if $q_U(f(v)) = q_V(v)$.
Direct sum: $(V, q_V) \oplus (U, q_U) = (V \oplus U, q_V \oplus q_U)$.
We say that (V, q) is *non-degenerate* if $(\, , \,)$ is.
For $L \subset V$ define $L^\perp = \{v \in V | \, (v, L) = 0\}$
L is *isotropic* if $L \subset L^\perp, \quad q|_L = 0$.
L is *Lagrangian* if $L = L^\perp, \quad q|_L = 0$.
(H, q) is *hyperbolic* if $\exists \, L \subset H$ Lagrangian.
$(V, q) \oplus (V, -q)$ is hyperbolic, with Lagrangian
$L = \{(v, v) | \, v \in V\}$.
Witt equivalence: $(V, q_V) \sim (U, q_U)$ if $\exists \, (H_i, q_i)$ hyperbolic
$(V, q_V) \oplus (H_1, q_1) \cong (U, q_U) \oplus (H_i, q_i)$ an isometry.
Witt group $W(F)$
Fact:

Quadratic vector spaces

Let V be a finite dimensional vector space over a field F
A function $q : V \to F$ is *quadratic* if $q(-x) = q(x)$ and
$(x, y) = q(x + y) - q(x) - q(y)$ is bilinear.
A linear map $f : V \to U$ is *isometry* if $q_U(f(v)) = q_V(v)$.
Direct sum: $(V, q_V) \oplus (U, q_U) = (V \oplus U, q_V \oplus q_U)$.
We say that (V, q) is *non-degenerate* if $(\, , \,)$ is.
For $L \subset V$ define $L^\perp = \{v \in V | \, (v, L) = 0\}$
L is *isotropic* if $L \subset L^\perp, \quad q|_L = 0$.
L is *Lagrangian* if $L = L^\perp, \quad q|_L = 0$.
(H, q) is *hyperbolic* if $\exists \, L \subset H$ Lagrangian.
$(V, q) \oplus (V, -q)$ is hyperbolic, with Lagrangian
$L = \{(v, v) | \, v \in V\}$.
Witt equivalence: $(V, q_V) \sim (U, q_U)$ if $\exists \, (H_i, q_i)$ hyperbolic
$(V, q_V) \oplus (H_1, q_1) \cong (U, q_U) \oplus (H_i, q_i)$ an isometry.
Witt group $W(F)$
Fact: unique *completely anisotropic* representative in each Witt class.

Quadratic abelian groups and Pointed categories

A is a finite abelian group, $q : A \to \mathbb{Q}/\mathbb{Z}$ a quadratic function.

Quadratic abelian groups and Pointed categories

A is a finite abelian group, $q : A \to \mathbb{Q}/\mathbb{Z}$ a quadratic function.
Witt group W_{pt}.

Quadratic abelian groups and Pointed categories

A is a finite abelian group, $q : A \to \mathbb{Q}/\mathbb{Z}$ a quadratic function.
Witt group W_{pt}.
Fact: $W_{pt} = \oplus_p W_{pt}(p)$, with $W_{pt}(p)$ finite 2-torsion of maximal order 8.

Quadratic abelian groups and Pointed categories

A is a finite abelian group, $q : A \to \mathbb{Q}/\mathbb{Z}$ a quadratic function.
Witt group W_{pt}.
Fact: $W_{pt} = \oplus_p W_{pt}(p)$, with $W_{pt}(p)$ finite 2-torsion of maximal order 8.
A fusion category is *pointed* if

Quadratic abelian groups and Pointed categories

A is a finite abelian group, $q : A \to \mathbb{Q}/\mathbb{Z}$ a quadratic function.
Witt group W_{pt}.
Fact: $W_{pt} = \oplus_p W_{pt}(p)$, with $W_{pt}(p)$ finite 2-torsion of maximal order 8.
A fusion category is *pointed* if all its simple objects are invertible.

Quadratic abelian groups and Pointed categories

A is a finite abelian group, $q : A \to \mathbb{Q}/\mathbb{Z}$ a quadratic function.
Witt group W_{pt}.
Fact: $W_{pt} = \oplus_p W_{pt}(p)$, with $W_{pt}(p)$ finite 2-torsion of maximal order 8.
A fusion category is *pointed* if all its simple objects are invertible.
Fact: the group A of isomorphism classes of simples of a braided pointed category is abelian.

Quadratic abelian groups and Pointed categories

A is a finite abelian group, $q : A \to \mathbb{Q}/\mathbb{Z}$ a quadratic function.
Witt group W_{pt}.
Fact: $W_{pt} = \oplus_p W_{pt}(p)$, with $W_{pt}(p)$ finite 2-torsion of maximal order 8.
A fusion category is *pointed* if all its simple objects are invertible.
Fact: the group A of isomorphism classes of simples of a braided pointed category is abelian.
Fact (Joyal-Street): $q : A \to \mathbb{Q}/\mathbb{Z}$ defined by $c_{a,a} = e^{2\pi i q(a)} 1_{aa}$ is a quadratic function,

Quadratic abelian groups and Pointed categories

A is a finite abelian group, $q : A \to \mathbb{Q}/\mathbb{Z}$ a quadratic function.
Witt group W_{pt}.
Fact: $W_{pt} = \oplus_p W_{pt}(p)$, with $W_{pt}(p)$ finite 2-torsion of maximal order 8.
A fusion category is *pointed* if all its simple objects are invertible.
Fact: the group A of isomorphism classes of simples of a braided pointed category is abelian.
Fact (Joyal-Street): $q : A \to \mathbb{Q}/\mathbb{Z}$ defined by $c_{a,a} = e^{2\pi i q(a)} 1_{aa}$ is a quadratic function, (A, q) (up to isometry) is a complete invariant (up to braided equivalences) of the braided pointed category.

Quadratic abelian groups and Pointed categories

A is a finite abelian group, $q : A \to \mathbb{Q}/\mathbb{Z}$ a quadratic function.
Witt group W_{pt}.
Fact: $W_{pt} = \oplus_p W_{pt}(p)$, with $W_{pt}(p)$ finite 2-torsion of maximal order 8.
A fusion category is *pointed* if all its simple objects are invertible.
Fact: the group A of isomorphism classes of simples of a braided pointed category is abelian.
Fact (Joyal-Street): $q : A \to \mathbb{Q}/\mathbb{Z}$ defined by $c_{a,a} = e^{2\pi i q(a)} 1_{aa}$ is a quadratic function, (A, q) (up to isometry) is a complete invariant (up to braided equivalences) of the braided pointed category.
Fact: Isotropic subgroups in (A, q) $\overset{1-1}{\longleftrightarrow}$ connected étale algebras in $\mathcal{C}(A, q)$.

Quadratic abelian groups and Pointed categories

A is a finite abelian group, $q : A \to \mathbb{Q}/\mathbb{Z}$ a quadratic function.
Witt group W_{pt}.
Fact: $W_{pt} = \oplus_p W_{pt}(p)$, with $W_{pt}(p)$ finite 2-torsion of maximal order 8.
A fusion category is *pointed* if all its simple objects are invertible.
Fact: the group A of isomorphism classes of simples of a braided pointed category is abelian.
Fact (Joyal-Street): $q : A \to \mathbb{Q}/\mathbb{Z}$ defined by $c_{a,a} = e^{2\pi i q(a)} 1_{aa}$ is a quadratic function, (A, q) (up to isometry) is a complete invariant (up to braided equivalences) of the braided pointed category.
Fact: Isotropic subgroups in (A, q) $\overset{1-1}{\longleftrightarrow}$ connected étale algebras in $\mathcal{C}(A, q)$.
Fact: (H, q) is hyperbolic iff $\mathcal{C}(H, q) \simeq \mathcal{Z}(\mathcal{S})$.

Non-degenerate braided categories

A braided fusion category $\mathcal{C}$ is *non-degenerate* if

Non-degenerate braided categories

A braided fusion category $\mathcal{C}$ is *non-degenerate* if
$\mathcal{Z}_{sym}(\mathcal{C}) = \{X \in \mathcal{C} |\ c_{X,Y}c_{Y,X} = 1_{Y\otimes X},\ \ Y \in \mathcal{C}\} \simeq \mathcal{V}ect$;

Non-degenerate braided categories

A braided fusion category $\mathcal{C}$ is *non-degenerate* if
$\mathcal{Z}_{sym}(\mathcal{C}) = \{X \in \mathcal{C} |\ c_{X,Y}c_{Y,X} = 1_{Y\otimes X},\ \ Y \in \mathcal{C}\} \simeq \mathcal{V}ect$;
or $\mathcal{C} \boxtimes \overline{\mathcal{C}} \to \mathcal{Z}(\mathcal{C})$ is an equivalence (Müger, Bruguières).

Non-degenerate braided categories

A braided fusion category $\mathcal{C}$ is *non-degenerate* if
$\mathcal{Z}_{sym}(\mathcal{C}) = \{X \in \mathcal{C} |\ c_{X,Y}c_{Y,X} = 1_{Y\otimes X},\ \ Y \in \mathcal{C}\} \simeq \mathcal{V}ect$;
or $\mathcal{C} \boxtimes \overline{\mathcal{C}} \to \mathcal{Z}(\mathcal{C})$ is an equivalence (Müger, Bruguières).
e.g. $\mathcal{C}(A, q)$ is non-degenerate iff q is;

Non-degenerate braided categories

A braided fusion category $\mathcal{C}$ is *non-degenerate* if
$\mathcal{Z}_{sym}(\mathcal{C}) = \{X \in \mathcal{C} |\ c_{X,Y}c_{Y,X} = 1_{Y\otimes X},\ \ Y \in \mathcal{C}\} \simeq \mathcal{V}ect$;
or $\mathcal{C} \boxtimes \overline{\mathcal{C}} \to \mathcal{Z}(\mathcal{C})$ is an equivalence (Müger, Bruguières).
e.g. $\mathcal{C}(A, q)$ is non-degenerate iff q is;
a modular tensor category is non-degenerate (Müger);

Non-degenerate braided categories

A braided fusion category $\mathcal{C}$ is *non-degenerate* if
$\mathcal{Z}_{sym}(\mathcal{C}) = \{X \in \mathcal{C} |\ c_{X,Y}c_{Y,X} = 1_{Y\otimes X},\ \ Y \in \mathcal{C}\} \simeq \mathcal{V}ect$;
or $\mathcal{C} \boxtimes \overline{\mathcal{C}} \to \mathcal{Z}(\mathcal{C})$ is an equivalence (Müger, Bruguières).
e.g. $\mathcal{C}(A, q)$ is non-degenerate iff q is;
a modular tensor category is non-degenerate (Müger);
$\mathcal{Z}(\mathcal{S})$ is non-degenerate for a (multi-)fusion $\mathcal{S}$ (DGNO).

Non-degenerate braided categories

A braided fusion category $\mathcal{C}$ is *non-degenerate* if
$\mathcal{Z}_{sym}(\mathcal{C}) = \{X \in \mathcal{C} |\ c_{X,Y}c_{Y,X} = 1_{Y\otimes X},\ \ Y \in \mathcal{C}\} \simeq \mathcal{V}ect$;
or $\mathcal{C} \boxtimes \overline{\mathcal{C}} \to \mathcal{Z}(\mathcal{C})$ is an equivalence (Müger, Bruguières).
e.g. $\mathcal{C}(A, q)$ is non-degenerate iff q is;
a modular tensor category is non-degenerate (Müger);
$\mathcal{Z}(\mathcal{S})$ is non-degenerate for a (multi-)fusion $\mathcal{S}$ (DGNO).
Witt equivalence: $\mathcal{C} \sim \mathcal{D}$ iff

Non-degenerate braided categories

A braided fusion category $\mathcal{C}$ is *non-degenerate* if
$\mathcal{Z}_{sym}(\mathcal{C}) = \{X \in \mathcal{C} |\ c_{X,Y}c_{Y,X} = 1_{Y\otimes X},\ \ Y \in \mathcal{C}\} \simeq \mathcal{V}ect$;
or $\mathcal{C} \boxtimes \overline{\mathcal{C}} \to \mathcal{Z}(\mathcal{C})$ is an equivalence (Müger, Bruguières).
e.g. $\mathcal{C}(A, q)$ is non-degenerate iff q is;
a modular tensor category is non-degenerate (Müger);
$\mathcal{Z}(\mathcal{S})$ is non-degenerate for a (multi-)fusion $\mathcal{S}$ (DGNO).
Witt equivalence: $\mathcal{C} \sim \mathcal{D}$ iff $\exists\ \mathcal{S}, \mathcal{T}$ and a braided equivalence
$\mathcal{C} \boxtimes \mathcal{Z}(\mathcal{S}) \simeq \mathcal{D} \boxtimes \mathcal{Z}(\mathcal{T})$.
Fact: for a connected étale algebra A in a non-degenerate category $\mathcal{C}$ $\quad \mathcal{Z}(\mathcal{C}_A) \simeq \mathcal{C} \boxtimes \overline{\mathcal{C}_A^{loc}}$.

Non-degenerate braided categories

A braided fusion category $\mathcal{C}$ is *non-degenerate* if
$\mathcal{Z}_{sym}(\mathcal{C}) = \{X \in \mathcal{C} |\ c_{X,Y}c_{Y,X} = 1_{Y\otimes X},\ \ Y \in \mathcal{C}\} \simeq \mathcal{V}ect$;
or $\mathcal{C} \boxtimes \overline{\mathcal{C}} \to \mathcal{Z}(\mathcal{C})$ is an equivalence (Müger, Bruguières).
e.g. $\mathcal{C}(A, q)$ is non-degenerate iff q is;
a modular tensor category is non-degenerate (Müger);
$\mathcal{Z}(\mathcal{S})$ is non-degenerate for a (multi-)fusion $\mathcal{S}$ (DGNO).
Witt equivalence: $\mathcal{C} \sim \mathcal{D}$ iff $\exists\ \mathcal{S}, \mathcal{T}$ and a braided equivalence
$\mathcal{C} \boxtimes \mathcal{Z}(\mathcal{S}) \simeq \mathcal{D} \boxtimes \mathcal{Z}(\mathcal{T})$.
Fact: for a connected étale algebra A in a non-degenerate category $\mathcal{C}$ $\quad \mathcal{Z}(\mathcal{C}_A) \simeq \mathcal{C} \boxtimes \overline{\mathcal{C}_A^{loc}}$. In particular $\mathcal{C} \sim \mathcal{C}_A^{loc}$.

Non-degenerate braided categories

A braided fusion category $\mathcal{C}$ is *non-degenerate* if
$\mathcal{Z}_{sym}(\mathcal{C}) = \{X \in \mathcal{C} |\ c_{X,Y}c_{Y,X} = 1_{Y\otimes X},\ \ Y \in \mathcal{C}\} \simeq \mathcal{V}ect$;
or $\mathcal{C} \boxtimes \overline{\mathcal{C}} \to \mathcal{Z}(\mathcal{C})$ is an equivalence (Müger, Bruguières).
e.g. $\mathcal{C}(A, q)$ is non-degenerate iff q is;
a modular tensor category is non-degenerate (Müger);
$\mathcal{Z}(\mathcal{S})$ is non-degenerate for a (multi-)fusion $\mathcal{S}$ (DGNO).
Witt equivalence: $\mathcal{C} \sim \mathcal{D}$ iff $\exists\ \mathcal{S}, \mathcal{T}$ and a braided equivalence
$\mathcal{C} \boxtimes \mathcal{Z}(\mathcal{S}) \simeq \mathcal{D} \boxtimes \mathcal{Z}(\mathcal{T})$.
Fact: for a connected étale algebra A in a non-degenerate category $\mathcal{C}$ $\quad \mathcal{Z}(\mathcal{C}_A) \simeq \mathcal{C} \boxtimes \overline{\mathcal{C}_A^{loc}}$. In particular $\mathcal{C} \sim \mathcal{C}_A^{loc}$.
Facts (DMNO): Witt trivial = monoidal centre ($\mathcal{Z}(\mathcal{S})$);

Non-degenerate braided categories

A braided fusion category $\mathcal{C}$ is *non-degenerate* if
$\mathcal{Z}_{sym}(\mathcal{C}) = \{X \in \mathcal{C} |\ c_{X,Y}c_{Y,X} = 1_{Y\otimes X},\ \ Y \in \mathcal{C}\} \simeq \mathcal{V}ect$;
or $\mathcal{C} \boxtimes \overline{\mathcal{C}} \to \mathcal{Z}(\mathcal{C})$ is an equivalence (Müger, Bruguières).
e.g. $\mathcal{C}(A, q)$ is non-degenerate iff q is;
a modular tensor category is non-degenerate (Müger);
$\mathcal{Z}(\mathcal{S})$ is non-degenerate for a (multi-)fusion $\mathcal{S}$ (DGNO).
Witt equivalence: $\mathcal{C} \sim \mathcal{D}$ iff $\exists\ \mathcal{S}, \mathcal{T}$ and a braided equivalence
$\mathcal{C} \boxtimes \mathcal{Z}(\mathcal{S}) \simeq \mathcal{D} \boxtimes \mathcal{Z}(\mathcal{T})$.
Fact: for a connected étale algebra A in a non-degenerate category $\mathcal{C}$ $\quad \mathcal{Z}(\mathcal{C}_A) \simeq \mathcal{C} \boxtimes \overline{\mathcal{C}_A^{loc}}$. In particular $\mathcal{C} \sim \mathcal{C}_A^{loc}$.
Facts (DMNO): Witt trivial = monoidal centre ($\mathcal{Z}(\mathcal{S})$);
In every Witt class there is unique (up to braided equivalence) completely anisotropic representative (*complete anisotropy* = no étale algebras).

Non-degenerate braided categories

A braided fusion category $\mathcal{C}$ is *non-degenerate* if
$\mathcal{Z}_{sym}(\mathcal{C}) = \{X \in \mathcal{C} |\ c_{X,Y}c_{Y,X} = 1_{Y\otimes X},\ \ Y \in \mathcal{C}\} \simeq \mathcal{V}ect$;
or $\mathcal{C} \boxtimes \overline{\mathcal{C}} \to \mathcal{Z}(\mathcal{C})$ is an equivalence (Müger, Bruguières).
e.g. $\mathcal{C}(A, q)$ is non-degenerate iff q is;
a modular tensor category is non-degenerate (Müger);
$\mathcal{Z}(\mathcal{S})$ is non-degenerate for a (multi-)fusion $\mathcal{S}$ (DGNO).
Witt equivalence: $\mathcal{C} \sim \mathcal{D}$ iff $\exists\ \mathcal{S}, \mathcal{T}$ and a braided equivalence
$\mathcal{C} \boxtimes \mathcal{Z}(\mathcal{S}) \simeq \mathcal{D} \boxtimes \mathcal{Z}(\mathcal{T})$.
Fact: for a connected étale algebra A in a non-degenerate category $\mathcal{C}$ $\quad \mathcal{Z}(\mathcal{C}_A) \simeq \mathcal{C} \boxtimes \overline{\mathcal{C}_A^{loc}}$. In particular $\mathcal{C} \sim \mathcal{C}_A^{loc}$.
Facts (DMNO): Witt trivial = monoidal centre ($\mathcal{Z}(\mathcal{S})$);
In every Witt class there is unique (up to braided equivalence) completely anisotropic representative (*complete anisotropy* = no étale algebras).
Fact (DNO): $W_{pt} \subset W_{tors}$ the latter is a 2-group with maximal order 32.

Vertex operator algebras

Rational VOA V :

Vertex operator algebras

Rational VOA V : $\mathcal{R}ep(V)$ is modular;

Vertex operator algebras

Rational VOA V : $\mathcal{R}ep(V)$ is modular;
V is unitary $=$ $\mathcal{R}ep(V)$ is (pseudo-)unitary .

Vertex operator algebras

Rational VOA V : $\mathcal{R}ep(V)$ is modular;
V is unitary $=$ $\mathcal{R}ep(V)$ is (pseudo-)unitary .
e.g. lattice VOA V_L, $\quad \mathcal{R}ep(V_L) = \mathcal{C}(L^{\#}/L, q)$;

Vertex operator algebras

Rational VOA V : $\mathcal{R}ep(V)$ is modular;
V is unitary $=$ $\mathcal{R}ep(V)$ is (pseudo-)unitary .
e.g. lattice VOA V_L, $\quad \mathcal{R}ep(V_L) = \mathcal{C}(L^{\#}/L, q)$;
e.g. affine VOA $\mathfrak{g}_k$, $\quad \mathcal{R}ep(\mathfrak{g}_k) = \mathcal{C}(\mathfrak{g}, k)$ (Bakalov-Kirillov).

Vertex operator algebras

Rational VOA V : $\mathcal{R}ep(V)$ is modular;
V is unitary $=$ $\mathcal{R}ep(V)$ is (pseudo-)unitary .
e.g. lattice VOA V_L, $\quad \mathcal{R}ep(V_L) = \mathcal{C}(L^{\#}/L, q)$;
e.g. affine VOA $\mathfrak{g}_k$, $\quad \mathcal{R}ep(\mathfrak{g}_k) = \mathcal{C}(\mathfrak{g}, k)$ (Bakalov-Kirillov).
Algebras in $\mathcal{R}ep(V)$ (HKL, KO): V unitary

Vertex operator algebras

Rational VOA V : $\mathcal{R}ep(V)$ is modular;
V is unitary $=$ $\mathcal{R}ep(V)$ is (pseudo-)unitary .
e.g. lattice VOA V_L, $\quad \mathcal{R}ep(V_L) = \mathcal{C}(L^{\#}/L, q)$;
e.g. affine VOA $\mathfrak{g}_k$, $\quad \mathcal{R}ep(\mathfrak{g}_k) = \mathcal{C}(\mathfrak{g}, k)$ (Bakalov-Kirillov).
Algebras in $\mathcal{R}ep(V)$ (HKL, KO): V unitary
Connected étale algebras in $\mathcal{R}ep(V)$ $\overset{1-1}{\longleftrightarrow}$ VOA extensions $\tilde{V}$ of V

Vertex operator algebras

Rational VOA V : $\mathcal{R}ep(V)$ is modular;
V is unitary $=$ $\mathcal{R}ep(V)$ is (pseudo-)unitary .
e.g. lattice VOA V_L, $\quad \mathcal{R}ep(V_L) = \mathcal{C}(L^{\#}/L, q)$;
e.g. affine VOA $\mathfrak{g}_k$, $\quad \mathcal{R}ep(\mathfrak{g}_k) = \mathcal{C}(\mathfrak{g}, k)$ (Bakalov-Kirillov).
Algebras in $\mathcal{R}ep(V)$ (HKL, KO): V unitary
Connected étale algebras in $\mathcal{R}ep(V)$ $\overset{1-1}{\longleftrightarrow}$ VOA extensions $\tilde{V}$ of V with $\mathcal{R}ep(V)_A^{loc} = \mathcal{R}ep(\tilde{V})$.

Vertex operator algebras

Rational VOA V : $\mathcal{R}ep(V)$ is modular;
V is unitary $=$ $\mathcal{R}ep(V)$ is (pseudo-)unitary .
e.g. lattice VOA V_L, $\quad \mathcal{R}ep(V_L) = \mathcal{C}(L^{\#}/L, q)$;
e.g. affine VOA $\mathfrak{g}_k$, $\quad \mathcal{R}ep(\mathfrak{g}_k) = \mathcal{C}(\mathfrak{g}, k)$ (Bakalov-Kirillov).
Algebras in $\mathcal{R}ep(V)$ (HKL, KO): V unitary
Connected étale algebras in $\mathcal{R}ep(V)$ $\overset{1-1}{\longleftrightarrow}$ VOA extensions $\tilde{V}$ of V with $\mathcal{R}ep(V)_A^{loc} = \mathcal{R}ep(\tilde{V})$.
Witt equivalence: $V \sim W$

Vertex operator algebras

Rational VOA V : $\mathcal{R}ep(V)$ is modular;
V is unitary $=$ $\mathcal{R}ep(V)$ is (pseudo-)unitary .
e.g. lattice VOA V_L, $\quad \mathcal{R}ep(V_L) = \mathcal{C}(L^{\#}/L, q)$;
e.g. affine VOA $\mathfrak{g}_k$, $\quad \mathcal{R}ep(\mathfrak{g}_k) = \mathcal{C}(\mathfrak{g}, k)$ (Bakalov-Kirillov).
Algebras in $\mathcal{R}ep(V)$ (HKL, KO): V unitary
Connected étale algebras in $\mathcal{R}ep(V)$ $\overset{1-1}{\longleftrightarrow}$ VOA extensions $\tilde{V}$ of V with $\mathcal{R}ep(V)_A^{loc} = \mathcal{R}ep(\tilde{V})$.
Witt equivalence: $V \sim W$ if there exist holomorphic U_i and

$$\begin{tikzcd}
V_0 & & V_2 & & \cdots & & & & V_{2n} \\
& V_1 \arrow[ul] \arrow[ur] & & \cdots \arrow[ul] & & & V_{2n-1} \arrow[ul] \arrow[ur] & &
\end{tikzcd}$$

where $V_0 = V \otimes U_0$ and $V_{2n} = W \otimes U_1$.

Vertex operator algebras

Rational VOA V : $\mathcal{R}ep(V)$ is modular;
V is unitary $=$ $\mathcal{R}ep(V)$ is (pseudo-)unitary .
e.g. lattice VOA V_L, $\quad \mathcal{R}ep(V_L) = \mathcal{C}(L^{\#}/L, q)$;
e.g. affine VOA $\mathfrak{g}_k$, $\quad \mathcal{R}ep(\mathfrak{g}_k) = \mathcal{C}(\mathfrak{g}, k)$ (Bakalov-Kirillov).
Algebras in $\mathcal{R}ep(V)$ (HKL, KO): V unitary
Connected étale algebras in $\mathcal{R}ep(V)$ $\overset{1-1}{\longleftrightarrow}$ VOA extensions $\tilde{V}$ of V with $\mathcal{R}ep(V)_A^{loc} = \mathcal{R}ep(\tilde{V})$.
Witt equivalence: $V \sim W$ if there exist holomorphic U_i and

$$\begin{tikzcd}
V_0 & & V_2 & & \cdots & & & & V_{2n} \\
& V_1 \arrow[ul] \arrow[ur] & & \cdots \arrow[ul] & & & V_{2n-1} \arrow[ul] \arrow[ur] & &
\end{tikzcd}$$

where $V_0 = V \otimes U_0$ and $V_{2n} = W \otimes U_1$.
Witt trivial = sub-holomorphic;

Vertex operator algebras

Rational VOA V : $\mathcal{R}ep(V)$ is modular;
V is unitary $=$ $\mathcal{R}ep(V)$ is (pseudo-)unitary .
e.g. lattice VOA V_L, $\quad \mathcal{R}ep(V_L) = \mathcal{C}(L^{\#}/L, q)$;
e.g. affine VOA $\mathfrak{g}_k$, $\quad \mathcal{R}ep(\mathfrak{g}_k) = \mathcal{C}(\mathfrak{g}, k)$ (Bakalov-Kirillov).
Algebras in $\mathcal{R}ep(V)$ (HKL, KO): V unitary
Connected étale algebras in $\mathcal{R}ep(V)$ $\overset{1-1}{\longleftrightarrow}$ VOA extensions $\tilde{V}$ of V with $\mathcal{R}ep(V)_A^{loc} = \mathcal{R}ep(\tilde{V})$.
Witt equivalence: $V \sim W$ if there exist holomorphic U_i and

$$\begin{tikzcd}
V_0 & & V_2 & & \cdots & & & & V_{2n} \\
& V_1 \arrow[ul] \arrow[ur] & & \cdots \arrow[ul] & & & V_{2n-1} \arrow[ul] \arrow[ur] & &
\end{tikzcd}$$

where $V_0 = V \otimes U_0$ and $V_{2n} = W \otimes U_1$.
Witt trivial = sub-holomorphic;
maximal VOAs in a Witt class have the same category of modules;

Vertex operator algebras

Rational VOA V : $\mathcal{R}ep(V)$ is modular;
V is unitary $=$ $\mathcal{R}ep(V)$ is (pseudo-)unitary .
e.g. lattice VOA V_L, $\quad \mathcal{R}ep(V_L) = \mathcal{C}(L^{\#}/L, q)$;
e.g. affine VOA $\mathfrak{g}_k$, $\quad \mathcal{R}ep(\mathfrak{g}_k) = \mathcal{C}(\mathfrak{g}, k)$ (Bakalov-Kirillov).
Algebras in $\mathcal{R}ep(V)$ (HKL, KO): V unitary
Connected étale algebras in $\mathcal{R}ep(V)$ $\overset{1-1}{\longleftrightarrow}$ VOA extensions $\tilde{V}$ of V with $\mathcal{R}ep(V)_A^{loc} = \mathcal{R}ep(\tilde{V})$.
Witt equivalence: $V \sim W$ if there exist holomorphic U_i and

$$\begin{tikzcd}
V_0 & & V_2 & & \cdots & & & & V_{2n} \\
& V_1 \arrow[ul] \arrow[ur] & & \cdots \arrow[ul] & & & V_{2n-1} \arrow[ul] \arrow[ur] & &
\end{tikzcd}$$

where $V_0 = V \otimes U_0$ and $V_{2n} = W \otimes U_1$.
Witt trivial = sub-holomorphic;
maximal VOAs in a Witt class have the same category of modules;
Witt inverses exist for lattice and affine VOAs

Vertex operator algebras

Rational VOA V : $\mathcal{R}ep(V)$ is modular;
V is unitary $=$ $\mathcal{R}ep(V)$ is (pseudo-)unitary .
e.g. lattice VOA V_L, $\quad \mathcal{R}ep(V_L) = \mathcal{C}(L^{\#}/L, q)$;
e.g. affine VOA $\mathfrak{g}_k$, $\quad \mathcal{R}ep(\mathfrak{g}_k) = \mathcal{C}(\mathfrak{g}, k)$ (Bakalov-Kirillov).
Algebras in $\mathcal{R}ep(V)$ (HKL, KO): V unitary
Connected étale algebras in $\mathcal{R}ep(V)$ $\overset{1-1}{\longleftrightarrow}$ VOA extensions $\tilde{V}$ of V with $\mathcal{R}ep(V)_A^{loc} = \mathcal{R}ep(\tilde{V})$.
Witt equivalence: $V \sim W$ if there exist holomorphic U_i and

$$\begin{tikzcd}
V_0 & & V_2 & & \cdots & & & & V_{2n} \\
& V_1 \arrow[ul] \arrow[ur] & & \cdots \arrow[ul] & & & V_{2n-1} \arrow[ul] \arrow[ur] & &
\end{tikzcd}$$

where $V_0 = V \otimes U_0$ and $V_{2n} = W \otimes U_1$.
Witt trivial = sub-holomorphic;
maximal VOAs in a Witt class have the same category of modules;
Witt inverses exist for lattice and affine VOAs

Let us collect some specimens from this zoo. The oldest and most venerable are surely the current algebras – also known as Wess-Zumino-Witten [2] theories. These current algebras have various extended algebras (a notion we explain below). So far, all known extended algebras are related to orbifolds [3] of WZW theories by a subgroup of the center. Another venerable example of rational theories are the minimal models of BPZ [1] and FQS [4] (and their N=1 and N=2 generalizations). These are based on the chiral algebra of the (N=1,or N=2 super-) Virasoro algebra itself, and these have rather nontrivial extensions known as W-algebras and their generalizations, W_n-algebras [5]. In addition there are various species of parafermions [6]. Between 1984 and 1986 it was realized [7] [8] [9] that parafermions and the various discrete series could be obtained by the GKO coset construction [10]. Indeed, any coset construction based on two rational chiral algebras will define a rational conformal field theory. Finally whenever the chiral algebra has a discrete symmetry we can form an orbifold [3] theory [11].

Clearly, this zoo should be organized. By trying to formulate all these theories in a unified way, we are led to conjecture: *all* RCFT's are related to certain deformations of groups, this deformation can be described axiomatically or in terms of 3D Chern-Simons-Witten (CSW) gauge theory and is closely related to certain quantum groups. A cynical version of this conjecture would state that nothing new has been found since 1986, so we must be done.

Moore-Seiberg conjecture

Conjecture (MS):

Moore-Seiberg conjecture

Conjecture (MS): Witt group of unitary VOAs W_{VOA} is generated by classes of lattice and affine VOAs.

Moore-Seiberg conjecture

Conjecture (MS): Witt group of unitary VOAs W_{VOA} is generated by classes of lattice and affine VOAs.

Monomorphism $W_{VOA} \to W_{MTC}$ into the Witt group of (pseudo-)unitary non-degenerate categories.

Moore-Seiberg conjecture

Conjecture (MS): Witt group of unitary VOAs W_{VOA} is generated by classes of lattice and affine VOAs.

Monomorphism $W_{VOA} \to W_{MTC}$ into the Witt group of (pseudo-)unitary non-degenerate categories.

Conjecture: Witt group W_{MTC} is generated by classes of lattice and affine VOAs.

Variations

A braided fusion category $\mathcal{C}$ is *non-degenerate over* $\mathcal{E}$ if

Variations

A braided fusion category $\mathcal{C}$ is *non-degenerate over* $\mathcal{E}$ if $\mathcal{Z}_{sym}(\mathcal{C}) \simeq \mathcal{E}$,

Variations

A braided fusion category $\mathcal{C}$ is *non-degenerate over* $\mathcal{E}$ if $\mathcal{Z}_{sym}(\mathcal{C}) \simeq \mathcal{E}$,
Deligne product over $\mathcal{E}$:

Variations

A braided fusion category $\mathcal{C}$ is *non-degenerate over* $\mathcal{E}$ if
$\mathcal{Z}_{sym}(\mathcal{C}) \simeq \mathcal{E}$,
Deligne product over $\mathcal{E}$: $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{D} = (\mathcal{C} \boxtimes \mathcal{D})_A$, where $A \in \mathcal{E} \boxtimes \mathcal{E}$.

Variations

A braided fusion category $\mathcal{C}$ is *non-degenerate over* $\mathcal{E}$ if
$\mathcal{Z}_{sym}(\mathcal{C}) \simeq \mathcal{E}$,
Deligne product over $\mathcal{E}$: $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{D} = (\mathcal{C} \boxtimes \mathcal{D})_A$, where $A \in \mathcal{E} \boxtimes \mathcal{E}$.
Witt trivial:

Variations

A braided fusion category $\mathcal{C}$ is *non-degenerate over* $\mathcal{E}$ if
$\mathcal{Z}_{sym}(\mathcal{C}) \simeq \mathcal{E}$,
Deligne product over $\mathcal{E}$: $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{D} = (\mathcal{C} \boxtimes \mathcal{D})_A$, where $A \in \mathcal{E} \boxtimes \mathcal{E}$.
Witt trivial: $\mathcal{Z}(\mathcal{S}, \mathcal{E})$ for $\mathcal{E} \subset \mathcal{Z}(\mathcal{S})$, where

Variations

A braided fusion category $\mathcal{C}$ is *non-degenerate over* $\mathcal{E}$ if
$\mathcal{Z}_{sym}(\mathcal{C}) \simeq \mathcal{E}$,
Deligne product over $\mathcal{E}$: $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{D} = (\mathcal{C} \boxtimes \mathcal{D})_A$, where $A \in \mathcal{E} \boxtimes \mathcal{E}$.
Witt trivial: $\mathcal{Z}(\mathcal{S}, \mathcal{E})$ for $\mathcal{E} \subset \mathcal{Z}(\mathcal{S})$, where
$\mathcal{Z}(\mathcal{S}, \mathcal{E}) = \mathcal{C}_{\mathcal{Z}(\mathcal{S})}(\mathcal{E}) = \{X \in \mathcal{Z}(\mathcal{S}) |\ c_{X,Y} c_{Y,X} = 1,\ \forall Y \in \mathcal{E}\}$.

Variations

A braided fusion category $\mathcal{C}$ is *non-degenerate over* $\mathcal{E}$ if
$\mathcal{Z}_{sym}(\mathcal{C}) \simeq \mathcal{E}$,
Deligne product over $\mathcal{E}$: $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{D} = (\mathcal{C} \boxtimes \mathcal{D})_A$, where $A \in \mathcal{E} \boxtimes \mathcal{E}$.
Witt trivial: $\mathcal{Z}(\mathcal{S}, \mathcal{E})$ for $\mathcal{E} \subset \mathcal{Z}(\mathcal{S})$, where
$\mathcal{Z}(\mathcal{S}, \mathcal{E}) = \mathcal{C}_{\mathcal{Z}(\mathcal{S})}(\mathcal{E}) = \{X \in \mathcal{Z}(\mathcal{S}) |\ c_{X,Y} c_{Y,X} = 1,\ \forall Y \in \mathcal{E}\}$.
Witt equivalence:

Variations

A braided fusion category $\mathcal{C}$ is *non-degenerate over* $\mathcal{E}$ if
$\mathcal{Z}_{sym}(\mathcal{C}) \simeq \mathcal{E}$,
Deligne product over $\mathcal{E}$: $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{D} = (\mathcal{C} \boxtimes \mathcal{D})_A$, where $A \in \mathcal{E} \boxtimes \mathcal{E}$.
Witt trivial: $\mathcal{Z}(\mathcal{S}, \mathcal{E})$ for $\mathcal{E} \subset \mathcal{Z}(\mathcal{S})$, where
$\mathcal{Z}(\mathcal{S}, \mathcal{E}) = \mathcal{C}_{\mathcal{Z}(\mathcal{S})}(\mathcal{E}) = \{X \in \mathcal{Z}(\mathcal{S}) |\ c_{X,Y} c_{Y,X} = 1,\ \forall Y \in \mathcal{E}\}$.
Witt equivalence: $\mathcal{C} \sim_{\mathcal{E}} \mathcal{D}$ iff

Variations

A braided fusion category $\mathcal{C}$ is *non-degenerate over* $\mathcal{E}$ if $\mathcal{Z}_{sym}(\mathcal{C}) \simeq \mathcal{E}$,
Deligne product over $\mathcal{E}$: $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{D} = (\mathcal{C} \boxtimes \mathcal{D})_A$, where $A \in \mathcal{E} \boxtimes \mathcal{E}$.
Witt trivial: $\mathcal{Z}(\mathcal{S}, \mathcal{E})$ for $\mathcal{E} \subset \mathcal{Z}(\mathcal{S})$, where
$\mathcal{Z}(\mathcal{S}, \mathcal{E}) = \mathcal{C}_{\mathcal{Z}(\mathcal{S})}(\mathcal{E}) = \{X \in \mathcal{Z}(\mathcal{S}) |\ c_{X,Y} c_{Y,X} = 1,\ \forall Y \in \mathcal{E}\}$.
Witt equivalence: $\mathcal{C} \sim_{\mathcal{E}} \mathcal{D}$ iff $\exists\ \mathcal{S}, \mathcal{T}$ over $\mathcal{E}$ and a braided equivalence $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{Z}(\mathcal{S}, \mathcal{E}) \simeq \mathcal{D} \boxtimes_{\mathcal{E}} \mathcal{Z}(\mathcal{T}, \mathcal{E})$.

Variations

A braided fusion category $\mathcal{C}$ is *non-degenerate over* $\mathcal{E}$ if $\mathcal{Z}_{sym}(\mathcal{C}) \simeq \mathcal{E}$,
Deligne product over $\mathcal{E}$: $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{D} = (\mathcal{C} \boxtimes \mathcal{D})_A$, where $A \in \mathcal{E} \boxtimes \mathcal{E}$.
Witt trivial: $\mathcal{Z}(\mathcal{S}, \mathcal{E})$ for $\mathcal{E} \subset \mathcal{Z}(\mathcal{S})$, where
$\mathcal{Z}(\mathcal{S}, \mathcal{E}) = \mathcal{C}_{\mathcal{Z}(\mathcal{S})}(\mathcal{E}) = \{X \in \mathcal{Z}(\mathcal{S}) |\ c_{X,Y} c_{Y,X} = 1,\ \forall Y \in \mathcal{E}\}$.
Witt equivalence: $\mathcal{C} \sim_{\mathcal{E}} \mathcal{D}$ iff $\exists\ \mathcal{S}, \mathcal{T}$ over $\mathcal{E}$ and a braided equivalence $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{Z}(\mathcal{S}, \mathcal{E}) \simeq \mathcal{D} \boxtimes_{\mathcal{E}} \mathcal{Z}(\mathcal{T}, \mathcal{E})$.
Witt group $W(\mathcal{E})$

Variations

A braided fusion category $\mathcal{C}$ is *non-degenerate over* $\mathcal{E}$ if $\mathcal{Z}_{sym}(\mathcal{C}) \simeq \mathcal{E}$,
Deligne product over $\mathcal{E}$: $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{D} = (\mathcal{C} \boxtimes \mathcal{D})_A$, where $A \in \mathcal{E} \boxtimes \mathcal{E}$.
Witt trivial: $\mathcal{Z}(\mathcal{S}, \mathcal{E})$ for $\mathcal{E} \subset \mathcal{Z}(\mathcal{S})$, where
$\mathcal{Z}(\mathcal{S}, \mathcal{E}) = \mathcal{C}_{\mathcal{Z}(\mathcal{S})}(\mathcal{E}) = \{X \in \mathcal{Z}(\mathcal{S}) |\ c_{X,Y} c_{Y,X} = 1,\ \forall Y \in \mathcal{E}\}$.
Witt equivalence: $\mathcal{C} \sim_{\mathcal{E}} \mathcal{D}$ iff $\exists\ \mathcal{S}, \mathcal{T}$ over $\mathcal{E}$ and a braided equivalence $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{Z}(\mathcal{S}, \mathcal{E}) \simeq \mathcal{D} \boxtimes_{\mathcal{E}} \mathcal{Z}(\mathcal{T}, \mathcal{E})$.
Witt group $W(\mathcal{E})$
e.g. $W(\mathcal{V}ect) = W$,

Variations

A braided fusion category $\mathcal{C}$ is *non-degenerate over* $\mathcal{E}$ if $\mathcal{Z}_{sym}(\mathcal{C}) \simeq \mathcal{E}$,
Deligne product over $\mathcal{E}$: $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{D} = (\mathcal{C} \boxtimes \mathcal{D})_A$, where $A \in \mathcal{E} \boxtimes \mathcal{E}$.
Witt trivial: $\mathcal{Z}(\mathcal{S}, \mathcal{E})$ for $\mathcal{E} \subset \mathcal{Z}(\mathcal{S})$, where
$\mathcal{Z}(\mathcal{S}, \mathcal{E}) = \mathcal{C}_{\mathcal{Z}(\mathcal{S})}(\mathcal{E}) = \{X \in \mathcal{Z}(\mathcal{S}) |\ c_{X,Y} c_{Y,X} = 1,\ \forall Y \in \mathcal{E}\}$.
Witt equivalence: $\mathcal{C} \sim_{\mathcal{E}} \mathcal{D}$ iff $\exists\ \mathcal{S}, \mathcal{T}$ over $\mathcal{E}$ and a braided equivalence $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{Z}(\mathcal{S}, \mathcal{E}) \simeq \mathcal{D} \boxtimes_{\mathcal{E}} \mathcal{Z}(\mathcal{T}, \mathcal{E})$.
Witt group $W(\mathcal{E})$
e.g. $W(\mathcal{V}ect) = W$, $W(\mathcal{R}ep(G))$

Variations

A braided fusion category $\mathcal{C}$ is *non-degenerate over* $\mathcal{E}$ if $\mathcal{Z}_{sym}(\mathcal{C}) \simeq \mathcal{E}$,
Deligne product over $\mathcal{E}$: $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{D} = (\mathcal{C} \boxtimes \mathcal{D})_A$, where $A \in \mathcal{E} \boxtimes \mathcal{E}$.
Witt trivial: $\mathcal{Z}(\mathcal{S}, \mathcal{E})$ for $\mathcal{E} \subset \mathcal{Z}(\mathcal{S})$, where
$\mathcal{Z}(\mathcal{S}, \mathcal{E}) = \mathcal{C}_{\mathcal{Z}(\mathcal{S})}(\mathcal{E}) = \{X \in \mathcal{Z}(\mathcal{S}) |\ c_{X,Y} c_{Y,X} = 1,\ \forall Y \in \mathcal{E}\}$.
Witt equivalence: $\mathcal{C} \sim_{\mathcal{E}} \mathcal{D}$ iff $\exists\ \mathcal{S}, \mathcal{T}$ over $\mathcal{E}$ and a braided equivalence $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{Z}(\mathcal{S}, \mathcal{E}) \simeq \mathcal{D} \boxtimes_{\mathcal{E}} \mathcal{Z}(\mathcal{T}, \mathcal{E})$.
Witt group $W(\mathcal{E})$
e.g. $W(\mathcal{V}ect) = W$, $W(\mathcal{R}ep(G))$ is related to G actions on non-degenerate categories and crossed G-categories;

Variations

A braided fusion category $\mathcal{C}$ is *non-degenerate over* $\mathcal{E}$ if $\mathcal{Z}_{sym}(\mathcal{C}) \simeq \mathcal{E}$,
Deligne product over $\mathcal{E}$: $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{D} = (\mathcal{C} \boxtimes \mathcal{D})_A$, where $A \in \mathcal{E} \boxtimes \mathcal{E}$.
Witt trivial: $\mathcal{Z}(\mathcal{S}, \mathcal{E})$ for $\mathcal{E} \subset \mathcal{Z}(\mathcal{S})$, where
$\mathcal{Z}(\mathcal{S}, \mathcal{E}) = \mathcal{C}_{\mathcal{Z}(\mathcal{S})}(\mathcal{E}) = \{X \in \mathcal{Z}(\mathcal{S}) |\ c_{X,Y} c_{Y,X} = 1,\ \forall Y \in \mathcal{E}\}$.
Witt equivalence: $\mathcal{C} \sim_{\mathcal{E}} \mathcal{D}$ iff $\exists\ \mathcal{S}, \mathcal{T}$ over $\mathcal{E}$ and a braided equivalence $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{Z}(\mathcal{S}, \mathcal{E}) \simeq \mathcal{D} \boxtimes_{\mathcal{E}} \mathcal{Z}(\mathcal{T}, \mathcal{E})$.
Witt group $W(\mathcal{E})$
e.g. $W(\mathcal{V}ect) = W$, $W(\mathcal{R}ep(G))$ is related to G actions on non-degenerate categories and crossed G-categories;
$W(s\mathcal{V}ect) = sW$ (DNO),

Variations

A braided fusion category $\mathcal{C}$ is *non-degenerate over* $\mathcal{E}$ if $\mathcal{Z}_{sym}(\mathcal{C}) \simeq \mathcal{E}$,
Deligne product over $\mathcal{E}$: $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{D} = (\mathcal{C} \boxtimes \mathcal{D})_A$, where $A \in \mathcal{E} \boxtimes \mathcal{E}$.
Witt trivial: $\mathcal{Z}(\mathcal{S}, \mathcal{E})$ for $\mathcal{E} \subset \mathcal{Z}(\mathcal{S})$, where
$\mathcal{Z}(\mathcal{S}, \mathcal{E}) = \mathcal{C}_{\mathcal{Z}(\mathcal{S})}(\mathcal{E}) = \{X \in \mathcal{Z}(\mathcal{S}) |\ c_{X,Y} c_{Y,X} = 1,\ \forall Y \in \mathcal{E}\}$.
Witt equivalence: $\mathcal{C} \sim_{\mathcal{E}} \mathcal{D}$ iff $\exists\ \mathcal{S}, \mathcal{T}$ over $\mathcal{E}$ and a braided equivalence $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{Z}(\mathcal{S}, \mathcal{E}) \simeq \mathcal{D} \boxtimes_{\mathcal{E}} \mathcal{Z}(\mathcal{T}, \mathcal{E})$.
Witt group $W(\mathcal{E})$
e.g. $W(\mathcal{V}ect) = W$, $W(\mathcal{R}ep(G))$ is related to G actions on non-degenerate categories and crossed G-categories;
$W(s\mathcal{V}ect) = sW$ (DNO),
Functoriality:

Variations

A braided fusion category $\mathcal{C}$ is *non-degenerate over* $\mathcal{E}$ if $\mathcal{Z}_{sym}(\mathcal{C}) \simeq \mathcal{E}$,
Deligne product over $\mathcal{E}$: $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{D} = (\mathcal{C} \boxtimes \mathcal{D})_A$, where $A \in \mathcal{E} \boxtimes \mathcal{E}$.
Witt trivial: $\mathcal{Z}(\mathcal{S}, \mathcal{E})$ for $\mathcal{E} \subset \mathcal{Z}(\mathcal{S})$, where
$\mathcal{Z}(\mathcal{S}, \mathcal{E}) = \mathcal{C}_{\mathcal{Z}(\mathcal{S})}(\mathcal{E}) = \{X \in \mathcal{Z}(\mathcal{S}) |\ c_{X,Y} c_{Y,X} = 1,\ \forall Y \in \mathcal{E}\}$.
Witt equivalence: $\mathcal{C} \sim_{\mathcal{E}} \mathcal{D}$ iff $\exists\ \mathcal{S}, \mathcal{T}$ over $\mathcal{E}$ and a braided equivalence $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{Z}(\mathcal{S}, \mathcal{E}) \simeq \mathcal{D} \boxtimes_{\mathcal{E}} \mathcal{Z}(\mathcal{T}, \mathcal{E})$.
Witt group $W(\mathcal{E})$
e.g. $W(\mathcal{V}ect) = W$, $W(\mathcal{R}ep(G))$ is related to G actions on non-degenerate categories and crossed G-categories;
$W(s\mathcal{V}ect) = sW$ (DNO),
Functoriality: homomorphism $W(\mathcal{E}) \to W(\mathcal{F})$ for a symmetric tensor functor $\mathcal{E} \to \mathcal{F}$.

Variations

A braided fusion category $\mathcal{C}$ is *non-degenerate over* $\mathcal{E}$ if $\mathcal{Z}_{sym}(\mathcal{C}) \simeq \mathcal{E}$,
Deligne product over $\mathcal{E}$: $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{D} = (\mathcal{C} \boxtimes \mathcal{D})_A$, where $A \in \mathcal{E} \boxtimes \mathcal{E}$.
Witt trivial: $\mathcal{Z}(\mathcal{S}, \mathcal{E})$ for $\mathcal{E} \subset \mathcal{Z}(\mathcal{S})$, where
$\mathcal{Z}(\mathcal{S}, \mathcal{E}) = \mathcal{C}_{\mathcal{Z}(\mathcal{S})}(\mathcal{E}) = \{X \in \mathcal{Z}(\mathcal{S}) |\ c_{X,Y} c_{Y,X} = 1,\ \forall Y \in \mathcal{E}\}$.
Witt equivalence: $\mathcal{C} \sim_{\mathcal{E}} \mathcal{D}$ iff $\exists\ \mathcal{S}, \mathcal{T}$ over $\mathcal{E}$ and a braided equivalence $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{Z}(\mathcal{S}, \mathcal{E}) \simeq \mathcal{D} \boxtimes_{\mathcal{E}} \mathcal{Z}(\mathcal{T}, \mathcal{E})$.
Witt group $W(\mathcal{E})$
e.g. $W(\mathcal{V}ect) = W$, $W(\mathcal{R}ep(G))$ is related to G actions on non-degenerate categories and crossed G-categories;
$W(s\mathcal{V}ect) = sW$ (DNO),
Functoriality: homomorphism $W(\mathcal{E}) \to W(\mathcal{F})$ for a symmetric tensor functor $\mathcal{E} \to \mathcal{F}$.
e.g. $\mathcal{V}ect \to s\mathcal{V}ect$ gives $S : W \to sW$,

Variations

A braided fusion category $\mathcal{C}$ is *non-degenerate over* $\mathcal{E}$ if $\mathcal{Z}_{sym}(\mathcal{C}) \simeq \mathcal{E}$,
Deligne product over $\mathcal{E}$: $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{D} = (\mathcal{C} \boxtimes \mathcal{D})_A$, where $A \in \mathcal{E} \boxtimes \mathcal{E}$.
Witt trivial: $\mathcal{Z}(\mathcal{S}, \mathcal{E})$ for $\mathcal{E} \subset \mathcal{Z}(\mathcal{S})$, where
$\mathcal{Z}(\mathcal{S}, \mathcal{E}) = \mathcal{C}_{\mathcal{Z}(\mathcal{S})}(\mathcal{E}) = \{X \in \mathcal{Z}(\mathcal{S}) |\ c_{X,Y} c_{Y,X} = 1,\ \forall Y \in \mathcal{E}\}$.
Witt equivalence: $\mathcal{C} \sim_{\mathcal{E}} \mathcal{D}$ iff $\exists\ \mathcal{S}, \mathcal{T}$ over $\mathcal{E}$ and a braided equivalence $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{Z}(\mathcal{S}, \mathcal{E}) \simeq \mathcal{D} \boxtimes_{\mathcal{E}} \mathcal{Z}(\mathcal{T}, \mathcal{E})$.
Witt group $W(\mathcal{E})$
e.g. $W(\mathcal{V}ect) = W$, $W(\mathcal{R}ep(G))$ is related to G actions on non-degenerate categories and crossed G-categories;
$W(s\mathcal{V}ect) = sW$ (DNO),
Functoriality: homomorphism $W(\mathcal{E}) \to W(\mathcal{F})$ for a symmetric tensor functor $\mathcal{E} \to \mathcal{F}$.
e.g. $\mathcal{V}ect \to s\mathcal{V}ect$ gives $S : W \to sW$,
the kernel =

Variations

A braided fusion category $\mathcal{C}$ is *non-degenerate over* $\mathcal{E}$ if $\mathcal{Z}_{sym}(\mathcal{C}) \simeq \mathcal{E}$,
Deligne product over $\mathcal{E}$: $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{D} = (\mathcal{C} \boxtimes \mathcal{D})_A$, where $A \in \mathcal{E} \boxtimes \mathcal{E}$.
Witt trivial: $\mathcal{Z}(\mathcal{S}, \mathcal{E})$ for $\mathcal{E} \subset \mathcal{Z}(\mathcal{S})$, where
$\mathcal{Z}(\mathcal{S}, \mathcal{E}) = \mathcal{C}_{\mathcal{Z}(\mathcal{S})}(\mathcal{E}) = \{X \in \mathcal{Z}(\mathcal{S}) |\ c_{X,Y} c_{Y,X} = 1,\ \forall Y \in \mathcal{E}\}$.
Witt equivalence: $\mathcal{C} \sim_{\mathcal{E}} \mathcal{D}$ iff $\exists\ \mathcal{S}, \mathcal{T}$ over $\mathcal{E}$ and a braided equivalence $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{Z}(\mathcal{S}, \mathcal{E}) \simeq \mathcal{D} \boxtimes_{\mathcal{E}} \mathcal{Z}(\mathcal{T}, \mathcal{E})$.
Witt group $W(\mathcal{E})$
e.g. $W(\mathcal{V}ect) = W$, $W(\mathcal{R}ep(G))$ is related to G actions on non-degenerate categories and crossed G-categories;
$W(s\mathcal{V}ect) = sW$ (DNO),
Functoriality: homomorphism $W(\mathcal{E}) \to W(\mathcal{F})$ for a symmetric tensor functor $\mathcal{E} \to \mathcal{F}$.
e.g. $\mathcal{V}ect \to s\mathcal{V}ect$ gives $S : W \to sW$,
the kernel = $\mathbb{Z}/16\mathbb{Z}$,

Variations

A braided fusion category $\mathcal{C}$ is *non-degenerate over* $\mathcal{E}$ if $\mathcal{Z}_{sym}(\mathcal{C}) \simeq \mathcal{E}$,
Deligne product over $\mathcal{E}$: $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{D} = (\mathcal{C} \boxtimes \mathcal{D})_A$, where $A \in \mathcal{E} \boxtimes \mathcal{E}$.
Witt trivial: $\mathcal{Z}(\mathcal{S}, \mathcal{E})$ for $\mathcal{E} \subset \mathcal{Z}(\mathcal{S})$, where
$\mathcal{Z}(\mathcal{S}, \mathcal{E}) = \mathcal{C}_{\mathcal{Z}(\mathcal{S})}(\mathcal{E}) = \{X \in \mathcal{Z}(\mathcal{S}) |\ c_{X,Y} c_{Y,X} = 1,\ \forall Y \in \mathcal{E}\}$.
Witt equivalence: $\mathcal{C} \sim_{\mathcal{E}} \mathcal{D}$ iff $\exists\ \mathcal{S}, \mathcal{T}$ over $\mathcal{E}$ and a braided equivalence $\mathcal{C} \boxtimes_{\mathcal{E}} \mathcal{Z}(\mathcal{S}, \mathcal{E}) \simeq \mathcal{D} \boxtimes_{\mathcal{E}} \mathcal{Z}(\mathcal{T}, \mathcal{E})$.
Witt group $W(\mathcal{E})$
e.g. $W(\mathcal{V}ect) = W$, $W(\mathcal{R}ep(G))$ is related to G actions on non-degenerate categories and crossed G-categories;
$W(s\mathcal{V}ect) = sW$ (DNO),
Functoriality: homomorphism $W(\mathcal{E}) \to W(\mathcal{F})$ for a symmetric tensor functor $\mathcal{E} \to \mathcal{F}$.
e.g. $\mathcal{V}ect \to s\mathcal{V}ect$ gives $S : W \to sW$,
the kernel = $\mathbb{Z}/16\mathbb{Z}$, the cokernel = 0?

Example

Relations between classes of $\mathcal{C}(sl(2), k)$ in the Witt group:

Example

Relations between classes of $\mathcal{C}(sl(2), k)$ in the Witt group:
$[\mathcal{C}(sl(2),\, 1)]^8 = 1$,

Example

Relations between classes of $\mathcal{C}(sl(2), k)$ in the Witt group:
$[\mathcal{C}(sl(2),\, 1)]^8 = 1$,
$[\mathcal{C}(sl(2),\, 8)] = [\mathcal{C}(sl(2),\, 3)]^{-2}[\mathcal{C}(sl(2),\, 1)]^2$,

Example

Relations between classes of $\mathcal{C}(sl(2), k)$ in the Witt group:
$[\mathcal{C}(sl(2),\, 1)]^8 = 1,$
$[\mathcal{C}(sl(2),\, 8)] = [\mathcal{C}(sl(2),\, 3)]^{-2}[\mathcal{C}(sl(2),\, 1)]^2,$
$[\mathcal{C}(sl(2),\, 28)] = [\mathcal{C}(sl(2),\, 3)][\mathcal{C}(sl(2),\, 1)]^{-1},$

Example

Relations between classes of $\mathcal{C}(sl(2), k)$ in the Witt group:
$[\mathcal{C}(sl(2),\, 1)]^8 = 1,$
$[\mathcal{C}(sl(2),\, 8)] = [\mathcal{C}(sl(2),\, 3)]^{-2}[\mathcal{C}(sl(2),\, 1)]^2,$
$[\mathcal{C}(sl(2),\, 28)] = [\mathcal{C}(sl(2),\, 3)][\mathcal{C}(sl(2),\, 1)]^{-1},$
$[\mathcal{C}(sl(2),\, 4)]^4 = 1,$

Example

Relations between classes of $\mathcal{C}(sl(2), k)$ in the Witt group:
$[\mathcal{C}(sl(2),\, 1)]^8 = 1,$
$[\mathcal{C}(sl(2),\, 8)] = [\mathcal{C}(sl(2),\, 3)]^{-2}[\mathcal{C}(sl(2),\, 1)]^2,$
$[\mathcal{C}(sl(2),\, 28)] = [\mathcal{C}(sl(2),\, 3)][\mathcal{C}(sl(2),\, 1)]^{-1},$
$[\mathcal{C}(sl(2),\, 4)]^4 = 1,$
$[\mathcal{C}(sl(2),\, 2)]^{16} = 1,$

Example

Relations between classes of $\mathcal{C}(sl(2), k)$ in the Witt group:
$[\mathcal{C}(sl(2),\, 1)]^8 = 1,$
$[\mathcal{C}(sl(2),\, 8)] = [\mathcal{C}(sl(2),\, 3)]^{-2}[\mathcal{C}(sl(2),\, 1)]^2,$
$[\mathcal{C}(sl(2),\, 28)] = [\mathcal{C}(sl(2),\, 3)][\mathcal{C}(sl(2),\, 1)]^{-1},$
$[\mathcal{C}(sl(2),\, 4)]^4 = 1,$
$[\mathcal{C}(sl(2),\, 2)]^{16} = 1,$
$[\mathcal{C}(sl(2),\, 10)] = [\mathcal{C}(sl(2),\, 2)]^7,$

Example

Relations between classes of $\mathcal{C}(sl(2), k)$ in the Witt group:
$[\mathcal{C}(sl(2),\, 1)]^8 = 1,$
$[\mathcal{C}(sl(2),\, 8)] = [\mathcal{C}(sl(2),\, 3)]^{-2}[\mathcal{C}(sl(2),\, 1)]^2,$
$[\mathcal{C}(sl(2),\, 28)] = [\mathcal{C}(sl(2),\, 3)][\mathcal{C}(sl(2),\, 1)]^{-1},$
$[\mathcal{C}(sl(2),\, 4)]^4 = 1,$
$[\mathcal{C}(sl(2),\, 2)]^{16} = 1,$
$[\mathcal{C}(sl(2),\, 10)] = [\mathcal{C}(sl(2),\, 2)]^7,$
$[\mathcal{C}(sl(2),\, 6)]^2 = [\mathcal{C}(sl(2),\, 2)]^3$